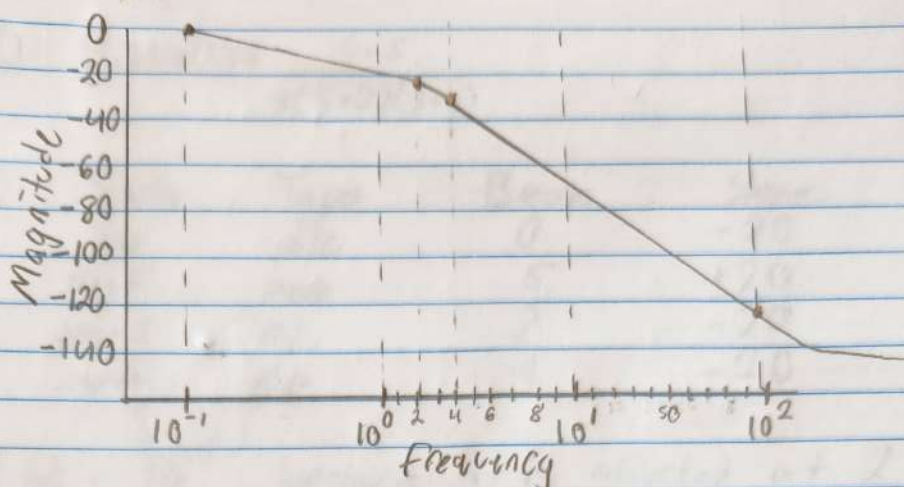


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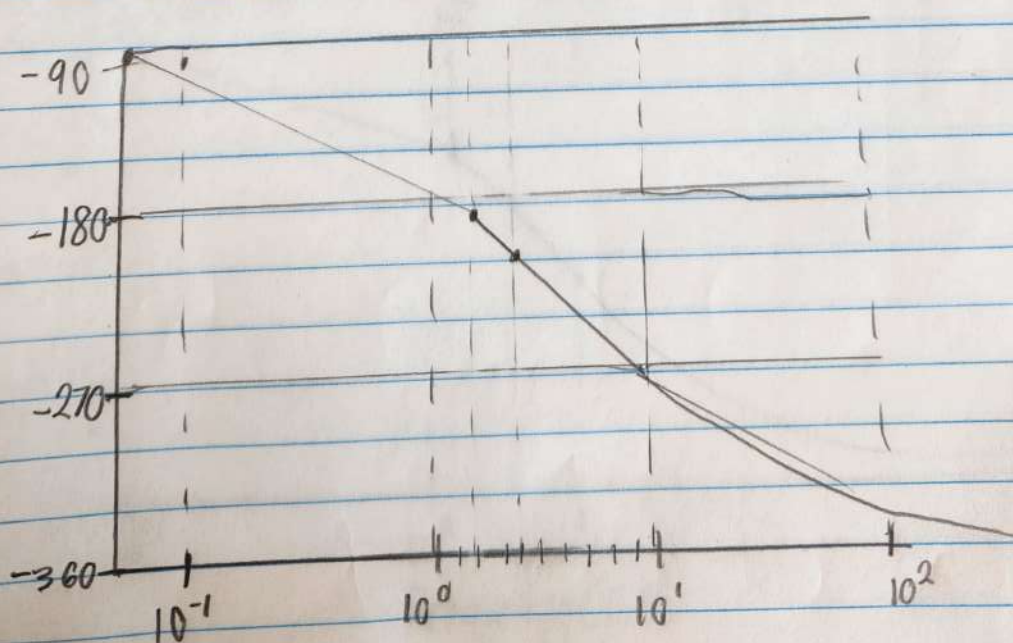


① $H(s)G(s) = \frac{1}{s(s+2)(s+4)}$

Factor	Type	Break Frequency	Slope
$1/s$	pole	0	-20
$1/(s+2)$	pole	2	-20
$1/(s+4)$	pole	4	-20

③ $\omega_i = 10^{-1}$; Because poles change it @ 2.

④ $|H(j\omega_i)G(j\omega_i)| = 1.25$; ⑤ $P_0 = -90^\circ, P_2 = -90^\circ, P_4 = -90^\circ$



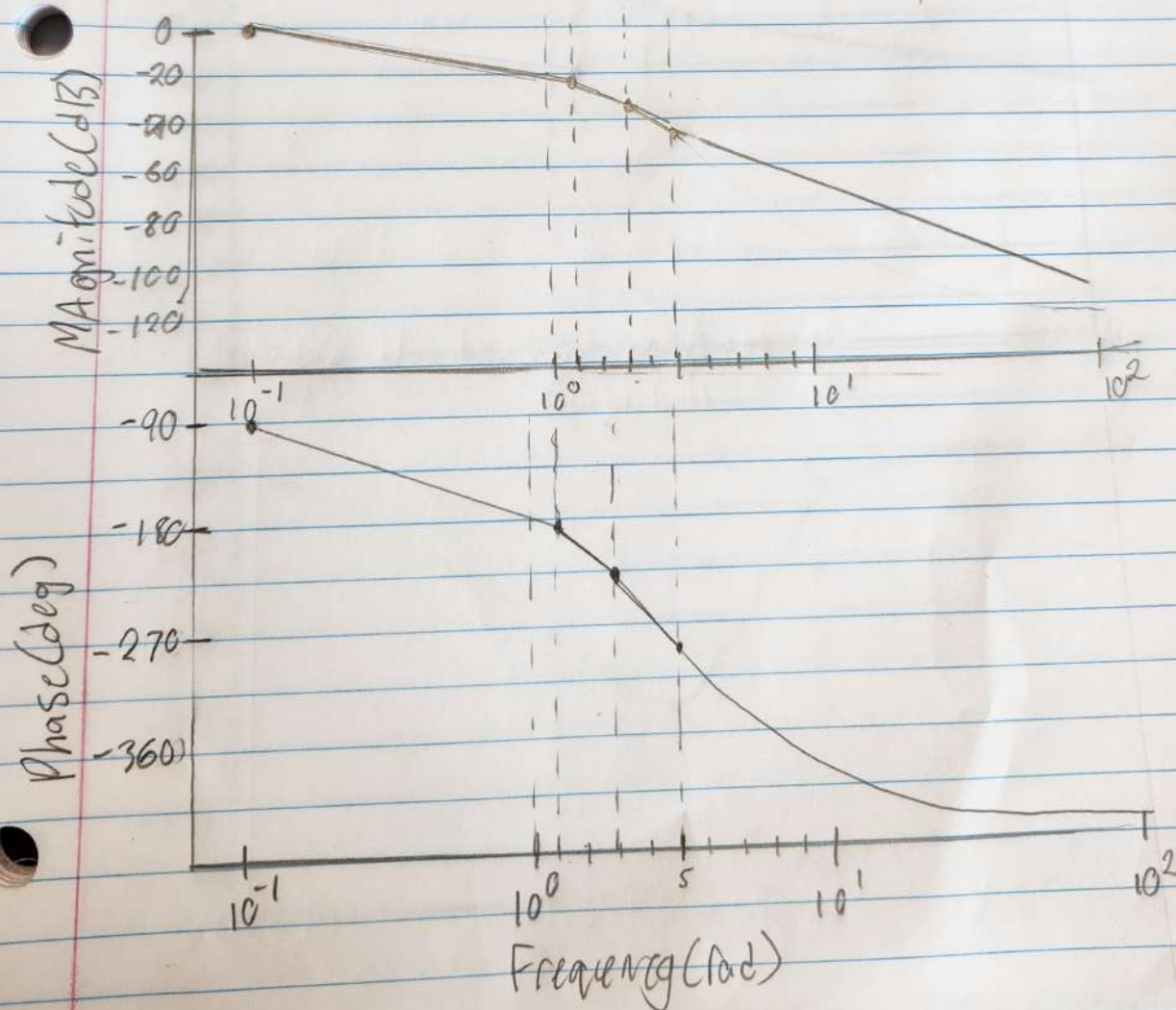
③ Bradley Schmidt
T007115861

① $H(s)G(s) = \frac{s+5}{s(s+2)(s+4)}$

Factor	Type	Break	Slope
s	pole	0	-20
$s+5$	zero	5	+20
$s+2$	pole	2	-20
$s+4$	pole	4	-20

② $\omega_i = 10^{-1}$ because it is affected at 2

④ 0.82



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15

① $G(s)H(s) = \frac{(s+5)(s+3)}{s(s+2)(s+4)}$

Factor	Type	Break	Slope
$1/s$	pole	0	-20
$1/(s+2)$	pole	2	-20
$1/(s+4)$	pole	4	-20
$s+5$	zero	5	+20
$s+3$	zero	3	+20

③ $20 \log |G(j\omega)| = 1.87$; ④ -91.24°

